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An Inquiry-Based Learning Approach to Mathematical Modeling

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Contents

Chapter 1

To the Instructor

1.1 Description of the Course

These notes have been used for three iterations of Math 246: Mathematical Modeling at Ripon College, a small liberal arts college. The course is a mid-level math course, with only Calculus 1 as a prerequisite. However, there are only a few places in the notes where derivatives are used, so that is not a strict content prerequisite; rather, it functions more as a "mathematical maturity" prerequisite. This course is not required of our majors, although most of them do take it, as do students from the natural sciences and economics. As such, the emphasis of the problems and projects in the notes is on the creation and interpretation of mathematical models, rather than on rigorous proofs or formal mathematics. The class size is typically 12-16 students, and we offer the course once every two years.

1.2 Role of Computation in the Notes

A departmental goal for this course is to give students an introduction to scientific computing and programming in Sage. The notes are designed for that language (which is based on Python), although the notes could be modified for your favorite programming language or computer algebra system. However, the notes emphasize computing above, say, proving theorems, so trying to use these notes without some computer package would not be advised.

In the most recent version of the course, students used upgraded accounts on CoCalc.com. This platform allows them to work in the cloud, so they do not have to download software. They can create Sage, Python, LaTeX documents and more in their accounts. The best feature of CoCalc.com for this course is the ease with which students can share documents with each other and with me. They can work simultaneously on single files, and they can submit their work directly through CoCalc.com for me to grade. There are free accounts available, but they can be unreliable, so I insist on the upgraded versions.

My students have typically had varying degrees of programming experience. In order to get them quickly up to speed on the programming ideas they will need for a chapter, I have "Sage Days" in advance of each chapter.

1.3 Structure of the Course/Grading

The main components of this course are class participation, #CS presentations, #PF presentations, writing assignments, projects, and skills checks. The grade breakdown is as follows:

- Class Participation: 10%
- #CS Presentations: 15%
- #PF Presentations: 5%
- Writing Assignments: 15%
- Projects: 25%
- Final Project: 15%
- Skills Checks: 15%

1.3.1 #CS and #PF presentations

I got the idea for the two types of presentations from Stan Yoshinobu's "The IBL Blog." Here is the post:
`http://theiblblog.blogspot.com/2014/06/productive-failure-pf.html`.

The #CS (Complete Solution) presentations are presentations of a solution to a problem. The focus of the presentation is to provide a clear explanation to the class of a best attempt at a correct solution. However, the solution may or may not be totally correct, and either is ok. At this point, we are emphasizing the process of doing and speaking about mathematics. Classmates will have the opportunity to ask questions and give suggestions to help correct or complete a solution.

The #CS presentations are graded on a scale of 0-4. Here is roughly what each grade means.

4: You gave a clear presentation of your ideas, and you had clearly put time into thinking deeply about the problem.

3: Your presentation was pretty clear, but there were places where it was a bit hard to follow. It was clear you thought about the problem.

2: Your presentation was either not very clear or it seemed like you needed to think more about the problem.

1: Your presentation was not prepared in advance and/or you did not attempt to solve the problem.

0: You did not present.

The #PF (Productive Failure) presentations are examples of a mistake a student made doing a problem and what he or she learned from that mistake. It is possible for a #PF to be a component of a #CS or for it to be a stand-alone presentation. In a #PF presentation, the student must present the mistake made AND explain what was learned from it. He or she might also give some explanation about how they realized the mistake.

The #PF presentations will be graded on a scale of 0-4. Here is roughly what the grades mean:

4: You gave a clear presentation of your mistake AND explained what you learned from it.

3: Either the presentation of your mistake or the presentation of what you learned from it could have been a bit more clear.

2: You gave only a presentation of your mistake or the presentation was not very clear.

1: You did not identify a clear mistake that you made.

0: You did not present.

Given my class sizes, students typically do 6-7 #CS presentations during the semester and are required to do 2 #PF presentations.

1.3.2 Writing Assignments

Writing assignments are given for every section. A typical writing assignment will consist of one-two problems of the students' choosing. Thus, writing assignments will differ from student to student! These solutions are graded for correctness, elegance and clarity. Writing assignments must be typed using LaTeXor submitted as Sage workbooks with explanations written in Markdown cells.

1.3.3 Projects

There are 4-5 projects throughout the semester that will allow the students apply the tools they learn to a modeling scenario. I have included a project with each chapter in the notes. Typically, these projects are done in pairs and require model creation, implementation, and analysis. The students write a formal mathematical paper in LaTeX detailing the results of their project.

Additionally, students do a final project at the end of the semester, for which they choose their own topic. They may choose to work with a partner or individually. They choose a problem that has not been extensively studied in class, create their own model and analyze it. They are required to use at least one technique we used in class, but they may also use additional mathematical techniques. The final project has both a written and presentation component.

1.3.4 Skills Checks

There are 3-4 skills checks throughout the semester. These are take-home assignments that take the place of traditional exams. I tried just doing projects at the end of the chapter, but I found that students needed a bit more incentive to really review all of the material they learned in a given chapter. I included my skills checks as an appendix to the notes.

1.4 Using the Notes

I have refined these notes over 3 times of teaching the course, and they have worked well for my students. Some of the problems are taken directly from the cited references, some are modified from those sources, and some are my own. I have included endnotes labeled "Note to instructor" to point out anything that might be useful to be aware of when using the notes. I have also included notes for the instructor after each project.

1.5 Acknowledgement

These notes were written with the generous support of the Academy of Inquiry-Based Learning and the Educational Advancement Foundation.

Chapter 2

Introduction to Mathematical Modeling

Note: I assign §1.1 in advance of the first day of class. We spend the first day engaging in group discussions and informal presentations of this material.

2.1 What is Mathematical Modeling?

A *mathematical model* is a model built using mathematical tools with the purpose of understanding real-world phenomena. The process of modeling, or creating a mathematical model, is a creative endeavor that will push you to use mathematical ideas from past courses as well as new ideas you will learn in this course or on your own.

Thought Question 1. *What are some math models you have previously studied? What real-world phenomena did they describe? What kinds of mathematical techniques or concepts were needed to understand the model?*

When we model a real-world system, some of the things we are interested in are the following:

- How does the system work?
- What causes change in the system?
- How sensitive is the system to certain changes?
- Can we predict what change will occur?
- Can we predict when change will occur?

Thought Question 2. *Think of one of the math models that you have previously studied. Try to answer as many of the questions above for your model.*

Although the modeling process is quite creative, there are some general steps that we will take in the construction of all of our models.

Definition 3. *We define the* modeling process *as follows:*

1. *Understand the problem or question being asked.*
2. *Make some simplifying assumptions.*

3. *Define all variables necessary for the model.*

4. *Construct a model.*

5. *Solve the problem(s) suggested by the mathematical model.*

6. *Validate and interpret the solution to your model within the context of the original problem.*

7. *Identify the strengths and weaknesses of the model.*

Problem 4. *Figure 2.1 is an aerial picture of a tree farm with each dot representing a tree. Your purpose is to provide an estimate for the lumber company about how many trees are in their lot so that they can make a revenue projection. Use the steps in Definition 3 to create a model, writing at least one sentence for each step.*

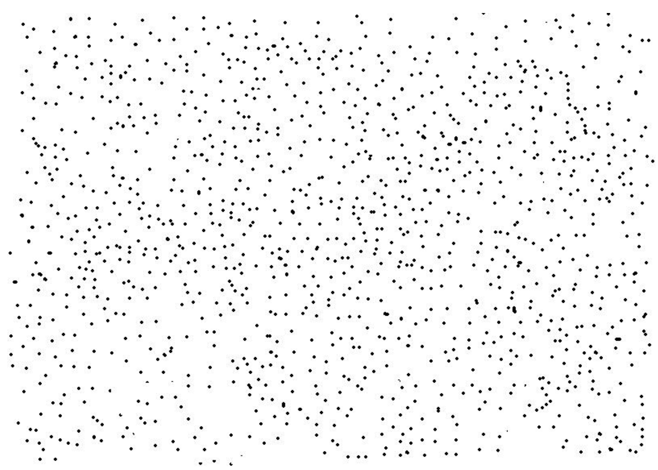


Figure 2.1: How many trees are in this lot?

2.2 Project: The Clepsydra

The clepsydra, or "water thief", measures time by the flow of water. It was used by the Egyptians, Greeks, and Romans to measure time. The clepsydra was used to determine the maximum length of speeches in the courts of justice and in the Roman senate. For the verbose among these orators, time literally ran out.

Figure 2.2: This water clock from the Western Han Dynasty of China (206 B.C. - A.D. 9) had a gauge with lines that marked divisions of time. As the water drained through the tube, the gauge sank, allowing the time to be read at each mark. Image taken from `https://www.metmuseum.org/art/collection/search/696219`.

In this project, you will model the behavior of a clepsydra. You have in front of you a homemade clepsydra. You can use water, a stopwatch, a ruler, a calculator, and any other tools you can find around the math department (other than the internet or books!!).

1. **How long to empty?**

   How long does it take for $\tilde{X}$ cm of water to drain from your clepsydra? (Here $\tilde{X}$ will be given to you at the very end of class, and you will not have the clepsydras to use any more.)

   Since you will not be able to experimentally determine this, you need to create a mathematical model. You should use Sage to create your model. You will submit a Sage notebook that contains your analysis. You should explain your work using the markdown cells. You should also submit a one-page description of your modeling process, using Definition 3 as a guide.

2. **How long to empty? (Calculus-based approach)** Assume you have a cylindrical bottle of radius R with a drilled hold in the bottom of radius r. For a positive number x, let $T(x)$ be the time in seconds for the bottle to empty after being filled to x cm above the bottom. The previous problem gave an experimental method to approximate the function T. Here is a different method.

   (a) Let h be a small increment in the height x of the water and consider the thin cylindrical section of water from x to $x+h$. Its potential energy above the bottom, *massgx*, is $\pi R^2 hgx$. When the water drains from height $x+h$ to height x a cylindrical section of water will appear beneath the hole that has the same volume. Show that its height is $(R/r)^2h$, so its kinetic energy is $1/2(R/r)^2hv(x)^2$.

(b) Use the fact that energy must be conserved to conclude Torricelli's law,

$$v(x) = \sqrt{2gx}.$$

(c) Use Torricelli's law and the definition of derivative to show that

$$T'(x) = (R^2/r^2\sqrt{2g})\sqrt{x}.$$

(d) Find $T(x)$ as the antiderivative of $T'(x)$ with the right constant.

(e) Compare the value of $T(\tilde{X})$ from problem 1 with that of this problem. Which seems more accurate? Why do you think that is?

3. **Challenge: Correct Shape (for students who have taken Calculus 2)**

Does your clepsydra have the property such that the water level drops the same amount vertically for equal time spans? If so, prove it.

If not, what shape should a clepsydra have if you want the water to drain in such a way that the water level drops equal vertical amounts in equal time intervals? Is the shape you found unique?

Your write-up should also include a detailed analysis of this question (including any assumptions), as well as any drawings that are useful.

Hint: You might find *Torricelli's law* from fluid dynamics useful in this problem. It says that the speed v of a fluid through a small hole at the bottom of a tank filled to a depth of h is the same as the speed that a body would obtain falling freely from the same height. In other words,

$$v = \sqrt{2gh},$$

where g is acceleration due to gravity.

2.3 Instructor's Notes for Clepsydra Project

The homemade clepsydra is an empty 2-liter bottle with a small hole drilled near the bottom and a ruler taped to the side. Students work on this project in groups of 3-4. Every group that I have had do this project has filled the clepsydra with water, timed how long it took to drain from various heights, and created a curve of best fit from this data. Since there is a lot of water involved in the data collection process, I have them do this project outside. I make my prediction for how long it will take $\tilde{X}$ cm of water to drain based on a theoretical model (assuming the shape of the clepsydra is perfectly cylindrical and using Torricelli's law as in problem 2). We then have a discussion about the relative effectiveness of our models.

We spend two 50-minute class periods on this project. The data collection takes them most of the first day, and then we have the second day to work on the Sage and LaTeXparts.

I typically don't assign the Challenge problem (part 2) because Calculus 2 is not a prerequisite for our course.

Chapter 3

Difference Equations

3.1 An Introduction to Difference Equations

We would like to model the behavior of dynamical systems, which can be thought of as a means of describing how one state develops into another state over time. In these models we will make the following assumption:

Future value = present value + change.

In other words, future states of the system depend explicitly on the past states.

In this chapter, we will study *difference equations*, which are recursively defined sequences. They can be used to model systems where change occurs incrementally rather than continuously. They can also be thought of as discrete approximations of continuous models (i.e. differential equations).[1]

Thought Question 5. *Can you think of an example of a real-world phenomena that would be better modeled by a difference equation than by a differential equation?*

A *time step* is the increment of time that passes between subsequent states. A time step could be any amount of time–an hour, a day, a year, a century–depending on what makes the most sense for your model. An *initial condition* is the value of your state when your model begins (often at time $t = 0$, but again, this choice of start time depends on your model).

Thought Question 6. *Give an example of a scenario that could be modeled by a difference equation with a time step of:*

- *an hour.*
- *a month.*
- *a year.*
- *a century.*

What would be a reasonable initial condition for each of your models?

Sage Problem 7. *You just won \$5000 in the Wisconsin lottery. Hooray! You decide to invest your winnings in an account that has an annual interest rate of 4%, compounded monthly.*

1. *Write a difference equation that represents this scenario. Let your time step be one month.*
2. *Using Sage, determine how much money you will have after 35 months.*
3. *How many months will it take for your account to reach \$10000?*

We will use *compartmental diagrams* as a way to pictorially represent our models. Compartmental diagrams are incredibly useful for both creating and explaining difference equations. They can also be used to create and study the more complicated discrete dynamical systems we will study later.

To create a compartmental diagram, draw a rectangle (or compartment) for each variable of interest and label the rectangle as such. Draw arrows to indicate the flow into or out of each variable. Label each arrow with the quantity that flows into or out of that variable during one time step. A labeled arrow pointing into the compartment indicates that the variable increases by that amount in each time step, whereas a labeled arrow pointing out of the compartment indicates that the variable decreases by that amount in each time step. Note: In this section, we are considering only systems of one variable, so we will have only one compartment in our diagrams. However, in subsequent sections, we will consider systems that have more than one variable and will draw diagrams with two or more compartments.

An example of a compartmental diagram for Problem 8 is given below.

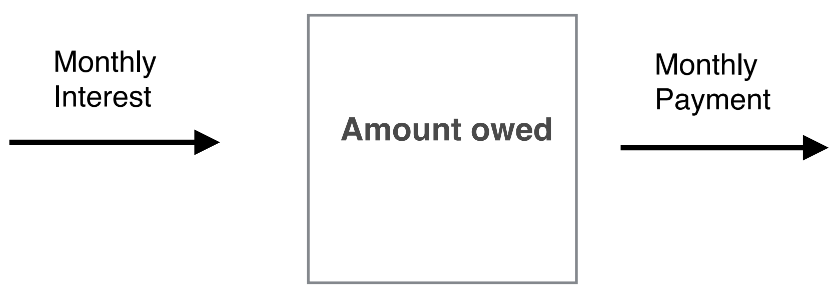


Figure 3.1: A compartmental diagram.

Sage Problem 8. *Suppose that you want to purchase a new car that costs \$25,000. You plan to make a down payment of \$2000, so you will need to finance \$23,000. You are able to negotiate a rate of 2% financing over a 72 month period. (Assume you pay 2% interest every month.) You decide to make a monthly payment of \$600.*

You would like to model the amount you still owe after n months.

1. *Use the compartmental diagram from Figure 3.1 to create a model.*
2. *Use Sage to plot your model on a relevant time interval.*
3. *How much do you still owe after 72 months?*

Problem 9. *Assume that Geometrania currently has 100 million people and that it has a population growth rate (birth rate minus death rate) of 1% per year. Also assume that it receives 100 thousand immigrants (new Geometranians) per year (who are quickly assimilated and reproduce at the same rate as the native population).*

1. *Draw a compartmental diagram for your model.*
2. *Determine the population 10 years from now.*

In general, difference equations, which are sometimes called recurrence relations, will have the form

$$a(n+1) = f(a(n), a(n-1), a(n-2), \ldots, a(0)). \tag{3.1}$$

The difference equation in Problem 8 is an example of a *first order non-homogeneous difference equation*. In general, these are any difference equations of the form

$$a(n+1) = ba(n) + C, \tag{3.2}$$

where $b, C \neq 0$. The "first order" bit refers to the value at a time step depending only on the value at one previous time step. The "non-homogeneous" bit refers to the addition of a nonzero constant C. A "homogeneous" difference equation would be one without the addition of a constant.

Thought Question 10. *What are some advantages of writing a difference equation recursively as in Equation 3.1? What are some disadvantages? (Hint: What if you want to know the value of a at time $t = k$ for k large?)*

It is sometimes possible to solve a difference equation in such a way as to avoid computing previous values. Such a solution is called a *closed form solution*.

Problem 11. *Consider the difference equation $a(n+1) = ra(n)$, where r is some fixed number and n is an integer. Find a closed form solution for a (i.e. an expression for $a(k)$ that is only in terms of r, k, and $a(0)$). What sort of behavior does this model have? How does the behavior of the model depend on r?*

Sage Problem 12. *Revisit Problem 7.*

1. *Find a closed form solution by hand, and use that to precisely answer Problem 7, question 3. You may try using the "find_root" command in Sage.*
2. *Plot your closed form solution using Sage.*

Problem 13. *Consider your model of financing a car in Problem 8. You would like to determine how much to pay each month so that your car is completely paid off at the end of 72 months.*

1. *Modify your model to allow for an arbitrary monthly payment. Call this monthly payment C.*
2. *Finding closed form solutions of non-homogeneous difference equations is more difficult than in the homogeneous case. The closed form solution to your model should be*

$$A(n) = (23000 - 50C) * \left(\frac{51}{50}\right)^n + 50C,$$

where $A(n)$ is the amount owed at the n^{th} month. Use this closed form solution to determine what your monthly payment should be to completely pay off the car at the end of 72 months.

3.2 Fixed Points and Stability

In this section, we will be interested in the following two questions:

1. What will be the long-time behavior of a system modeled by a difference equation?
2. Will the long-time behavior of the system depend on the initial condition?

Thought Question 14. *What is meant by the phrase* long-time behavior*? How would you express this idea using mathematical notation?*

Informally, when a dynamical system "settles down," the point to which it stabilizes is called a *fixed point*. The system is then said to be in *steady state* or equilibrium. To be more precise, we make the following definition.

Definition 15. *A* fixed point $\bar{a}$ *of the (first order) difference equation* $a(n+1) = f(a(n))$ *is a point such that* $f(\bar{a}) = \bar{a}$*.*

Thought Question 16. *Do all difference equations have fixed points? If so, say why. If not, provide a counterexample.*

Thought Question 17. *Do all scenarios modeled using difference equations have physically significant fixed points? Why or why not?*

Problem 18.

1. *Does your model in Problem 7 have any fixed points? Why or why not?*
2. *Find the fixed point(s) of your model in Problem 8. What is the physical interpretation of this value?*

Problem 19.

1. *Find the fixed point(s) of a general first order non-homogeneous difference equation (Eq. 3.2).*
2. *Will the long-time behavior of a first order non-homogeneous difference equation depend on its initial conditions? Why or why not?*

Once we know that $\bar{a}$ is a fixed point of a difference equation $a(n+1) = f(a(n))$, we would like to know whether the fixed point is *stable* or *unstable*. Roughly speaking, a stable fixed point is one such that small perturbations still gets sent under the difference equation towards the fixed point. An unstable fixed point is one such that any perturbation gets sent far away from the fixed point.

Thought Question 20. *Consider the difference equation* $x(n+1) = -.3x(n) + 2$*. What is the fixed point of this system? Start with a number close to the fixed point and iterate your system using a calculator or Sage. Do you think this fixed point is stable or unstable? Why?*

Here is a theorem that we will use without proof.

Theorem 21. *If $\bar{a}$ is a fixed point of a first order difference equation $a(n+1) = f(a(n))$, then $\bar{a}$ is a stable fixed point if $|f'(\bar{a})| < 1$. Similarly, $\bar{a}$ is unstable if $|f'(\bar{a})| > 1$.*

If $|f'(\bar{a})| < 1$, we call $\bar{a}$ an attracting fixed point. *If $|f'(\bar{a})| > 1$, we call $\bar{a}$ a* repelling fixed point.

Problem 22. *Give an explicit characterization of when fixed points of a first order nonhomogeneous difference equation (Equation 3.2) are stable.*

Problem 23. *In this problem, you will create a model of a crane population that includes "hacking" chicks, which means to hatch chicks in captivity and release them into the wild. Assume that your crane population has a constant birth rate per year of 2% of the population and a constant death rate of 2.5%. Assume also that 5 chicks are "hacked" each year; i.e. assume 5 hatched chicks are released into the wild and absorbed into the population.*

1. *Draw a compartmental diagram for your model.*
2. *Use the compartmental diagram to create your model.*
3. *What are the fixed points of your model? Are they stable or unstable? How can you interpret this physically?*

Problem 24. *Find the fixed points of the following difference equations and determine if they are stable or unstable.*

1. $x(n) = \frac{x(n-1)}{2+x(n-1)}$.
2. $x(n) = a \cdot x(n-1)e^{r \cdot x(n-1)}$, *where a and r are constants, and $a \neq 1$.*

Sage Problem 25 (The Logistic Model of Population Growth)**.**[2] *So far, we have seen one basic model of population growth; namely, the exponential growth equation. In the real world, there is limited application of such an equation to modeling a population (why?). In this problem, you will develop a population model with a variable growth rate, $r(x(n))$. In other words, you will find a model of the form*

$$x(n+1) = r(x(n))x(n) + x(n). \tag{3.3}$$

Suppose we are trying to model the growth of some population. We will make the following assumptions:

- *As a population increases, the growth rate declines.*
- *There is a population that cannot be exceeded called the* carrying capacity. *Denote the carrying capacity C. If a population is near the carrying capacity, its growth rate would be getting close to zero.*
- *If the population is small, the growth rate would be the largest. Call the growth rate at population zero the* intrinsic growth rate. *Denote the growth rate at population zero K.*

We will use these assumptions to derive the so-called logistic model.

1. *Are these assumptions reasonable? Why or why not?*
2. *Using the assumptions, find an expression for* $r(x(n))$*. (Hint: Assume that your growth rate function is linear. What are two points that it should pass through?)*
3. *Put your expression for* $r(x(n))$ *into Equation 3.3. This is the logistic model.*
4. *Use Sage to plot the behavior of this model for a population with a carrying capacity of 100 and initial population of 1. Use the values 0.5, 1.3, 2.2, and 2.7 for your intrinsic growth rates. What do you notice?*

Problem 26. *What are the fixed points of the logistic model? Are they stable or unstable? How can you interpret this observation?*

3.3 Linear Recurrence Relations with Constant Coefficients[3]

There are certain types of recurrence relations that one can solve by hand (i.e. without using Sage). One such recurrence relation is of the form

$$a_0x(n) + a_1x(n-1) + a_2x(n-2) + \cdots + a_mx(n-m) = b. \tag{3.4}$$

Here all of the a_i and b are constants. Such an equation is called an *mth-order linear recurrence relation with constant coefficients*. When $b = 0$, we say that the equation is *homogeneous*. In other words,

$$a_0x(n) + a_1x(n-1) + a_2x(n-2) + \cdots + a_mx(n-m) = 0 \tag{3.5}$$

is a homogeneous mth-order linear recurrence relation.

Thought Question 27. *Give an example of a first-order linear recurrence relation with constant coefficients. (Don't think too hard–we've seen at least one before!)*

Thought Question 28. *Why do we call Equations 3.4 and 3.5 linear? How does this relate to usages of that term that you may have seen in other courses?*

A simple example of a linear recurrence relation with constant coefficients that we have already seen is the exponential equation. Recall that this had the form

$$x(n) = Rx(n-1).$$

This equation had as its solution $x(n) = R^nx(0)$.

Problem 29. *What quantity determines the long-time behavior of the exponential equation? State explicitly the possibilities for the long-time behavior.*

In some sense, linear recurrence relations with constant coefficients are generalizations of the exponential equation. As such, you might be tempted to ask if we can find a solution that is similar to that of the exponential equation. In fact, that idea will guide our process to solve linear recurrence relations with constant coefficients.

Problem 30. *Suppose that both $c_1\lambda_1^n$ and $c_2\lambda_2^n$ are solutions to Equation 3.5. Show that $c_1\lambda_1^n + c_2\lambda_2^n$ is also a solution.*

Problem 31 (Fibonacci Sequence)**.** *The Fibonacci sequence is the sequence of numbers $1, 1, 2, 3, 5, 8, \ldots$.*

1. *Let $f(n)$ represent the nth term of the Fibonacci sequence. Write a recurrence relation for $f(n)$.*
2. *We would like to find a closed form solution for $f(n)$. Since we are being guided by what we know about the exponential equation, assume that your solution is of the form $f(n) = c\lambda^n$, where c and λ are numbers to be determined. Use your equation from 1 to solve for λ (you may obtain more than one).*
3. *Use Problem 30 to write a general solution for $f(n)$.*
4. *Solve for your constants using initial data.*
5. *Use your closed form solution to find the 232th term in the Fibonacci sequence.*

Problem 32. *In this problem, we will generalize the method developed in Problem 31 in order to solve Equation 3.5.*

1. *Assume that your solution has the form $x(n) = c\lambda^n$. Using Equation 3.5, what equation does λ have to satisfy? This equation is called the* characteristic equation.
2. *In general, how many possible λ does Equation 3.5 have? These λ are called the* eigenvalues.
3. *Write the general solution to Equation 3.5.*

Problem 33. *In Problem 32, you found a general solution to Equation 3.5. Write a paragraph describing the long-time behavior of the solution to a general second-order homogeneous recurrence relation with constant coefficients. (Hint: You might first consider the case of real eigenvalues. Then move onto the case where the eigenvalues are complex. Remember that a complex number $a + bi$ can be written as $re^{i\theta}$, where $r = \sqrt{a^2 + b^2}$ and $\theta = \arctan(b/a)$. Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$, might also be useful.)*

3.4 Systems of First Order Difference Equations

We will call a system of several discrete difference equations a *discrete dynamical system*. We will consider first order systems such as

$$\begin{aligned} x(n) &= f(x(n-1), y(n-1), z(n-1)) \\ y(n) &= g(x(n-1), y(n-1), z(n-1)) \\ z(n) &= h(x(n-1), y(n-1), z(n-1)), \end{aligned}$$

where f, g, h are some functions.

Problem 34. *1. State in mathematical notation what we should mean by a fixed point of the discrete dynamical system given above.*

2. Using your definition, find the fixed point(s) of the discrete dynamical system

$$\begin{aligned} x(n) &= x(n-1) - y(n-1) \\ y(n) &= 2y(n-1) + x(n-1). \end{aligned}$$

As in the case of a discrete difference equation, we can make sense of what it means for a fixed point to be stable or unstable. We will not make this fully precise, although the interested student might think about what would be needed to do so.

Suppose that you have a system of two first-order difference equations such as

$$\begin{aligned} a(n) &= F(a(n-1), b(n-1)) \\ b(n) &= G(a(n-1), b(n-1)), \end{aligned}$$

with fixed point $(\bar{a}, \bar{b})$. One way to study the stability of $(\bar{a}, \bar{b})$ is to draw a *phase plane portrait*. A phase plane portrait is a graph of a vs b.

Thought Question 35. *Why is plotting your solution in a phase plane portrait useful? What information can you gain?*

Sage Problem 36. *1. For the system of equations given in Problem 34, use Sage to draw a phase plane portrait. Make sure to include the fixed point. Would you say the fixed point is stable or unstable? Why?*

2. For the system of equations

$$\begin{aligned} x(n) &= .4x(n-1) - .2y(n-1) \\ y(n) &= .3x(n-1) + .3y(n-1), \end{aligned}$$

use Sage to draw a phase plane portrait, including the fixed point. Would you say the fixed point is stable or unstable? Why?

Sage Problem 37. ***Military insurgencies**: In this scenario, you will model the dynamics of military insurgency and police forces in Mathylvania. You can make the following assumptions:*

- *Approximately* 8% *of insurgents switch sides and join the police force each week.*
- *Approximately* 11% *of police switch sides and join the insurgency each week.*
- *Some fixed number of new insurgents arrive each week from the neighboring country of Recursionstan.*

- *Recruiting efforts in Mathylvania yield a fixed number of new police recruits each week.*
- *In armed conflict with insurgent forces, the local police are able to capture or kill approximately* 10% *of the insurgent force each week while losing about* 3% *of the police force each week.*

1. *Draw a compartmental diagram and use it to derive a system of discrete difference equations that display the desired behavior.*
2. *Plot the behaviors of the insurgent and police populations over 50 weeks. What behavior do you notice?*
3. *Find the fixed point of the system.*
4. *Create a phase plane portrait. How do you interpret this plot? Does the fixed point appear to be stable or unstable.*

Sage Problem 38. ***Star-crossed lovers****: In this scenario, you will model the love felt between Romeo and Juliet; namely, you would like to model the love Romeo feels for Juliet and the love that Juliet feels for Romeo. You can make the following assumptions:*

- *The love Juliet feels for Romeo depends on the love she feels for him and the love he feels for her.*
- *The love Romeo feels for Juliet depends on the love he feels for her and the love she feels for him.*

1. *Let $J(n)$ denote the amount of love Juliet has for Romeo at time n, and let $R(n)$ denote the amount of love Romeo has for Juliet at time n. A positive value of J or R indicates love, whereas a negative value indicates hate. A value of zero indicates a neutral feeling.*

   Assume your model is of the form

   $$\begin{aligned} J(n+1) &= aJ(n)+bR(n) \\ R(n+1) &= cJ(n)+dR(n), \end{aligned}$$

   where a,b,c,d are real numbers (not necessarily positive).

   What happens when a,b,c,d are all positive? (Explore different magnitudes of these constants.) How would you interpret the outcome(s)?
2. *What happens when a and d are positive, but b and c are negative? (Explore different magnitudes of these constants.) How would you interpret the outcome(s)?*

3.5 Predator-Prey Model of Population Dynamics

All of the models we have analyzed so far have been linear, except for the logistic model. Many (most?) real world scenarios are best modeled using non-linear systems. This section will focus on the predator-prey model of population dynamics. The classical model was posed by A.J. Lotka and V. Volterra in the 1920s. There are many generalizations and refinements of this model in the literature.

We will be interested in modeling the interaction between two species: a predator and its prey. Clearly, the population of one group will depend on that of the other.

Problem 39. *For an initial model of predator-prey dynamics, we make the following assumptions [14]:*

1. *There are two species that interact: a predator species and a prey species. There are no other species interacting with these two.*
2. *In the absence of the predator, the prey exhibits pure exponential growth.*
3. *In the absence of the prey, the predator species dies out exponentially.*
4. *The predators kill the prey in such a way that the predator population increases at a rate proportional to the product of the number of predators and the number of prey. The prey population decreases at a rate proportional to this product. (This assumption comes from* the law of mass action.*)* [4]

Using the assumptions above, create a system of two difference equations that model the population of the predator, $P(n)$, and that of the prey, $N(n)$, over time.

Thought Question 40. *What are some strengths and weaknesses of this model? What other assumptions are implied by assumptions 2 and 3?*

Sage Problem 41.

1. *Using the model you created in Problem 39, plot both populations versus time on the same set of axes. What do you notice?* [5]
2. *Plot a phase plane portrait of the two populations. How does the behavior of this graph relate to what you observed in Part 1?*

Now that we have several population models, we can combine them (like building blocks) to create more nuanced models.

Sage Problem 42.

1. *Modify the basic predator-prey model from Problem 39 so that, in the absence of the predator, the prey population follows a logistic model. Let the other assumptions remain the same.*
2. *Use Sage to plot both populations versus time on the same set of axes. What do you notice?*
3. *Plot a phase portrait of the two populations? How does the behavior of this graph relate to what you observed in Part 2?*
4. *Describe the qualitative differences between this model and the original predator-prey model.*

3.6 Project: Rumors, they are a-spreadin'

As students at a small liberal arts school, you are no doubt familiar with how quickly a rumor can spread among a population. In this project, we will use difference equations to model the spread of a rumor among our student body.

Suppose we have a particularly juicy rumor, like "Your professor has a tattoo." You will model how quickly this rumor will spread among the college community. To construct your model, you should consider (at least) three groups of people:

- Spreaders, who know the rumor and actively spread it when meeting another person,
- Ignorants, who do not know the rumor,
- Stiflers, who have heard the rumor, but are no longer interested in spreading it.

Using the modeling process (Definition 3), you should create a system of (at least) three difference equations to model this scenario. Some questions you might consider are:

- What assumptions should you make about the behavior of these groups?
- What percentage of the student body will hear the rumor after a certain time?
- What conditions does the previous question depend upon?
- Are there any strategies that could be employed to mitigate the spread of rumors?

You will submit a paper (written in LaTeX) with the details of your analysis. You will not turn in a separate Sage worksheet, so appropriate figures should be included in your paper.

3.7 Instructor's Notes for Rumor Project

The intention of this project is for students to use the machinery they have developed in this chapter to create their own model. Typically, the models they create are variations on the SIR model of disease spread applied to the context of rumors. It is important to emphasize that they use compartmental diagrams to design their model and to think about which terms in the model are best modeled using mass action terms.

Typically I give the students a week for this project. I employ a day of peer review for the papers, as this is the first time some of them are writing significant mathematics papers.

Chapter 4

Markov Chains

A *Markov chain* is a mathematical system that undergoes a transition from one state to another, with finite or countably many possibilities. A Markov chain is a random process such that the subsequent state depends only on the current state of the system, not on any states that came before. Markov chains can be used to model many real-world phenomena. The material for this chapter will be based on [11] and [13].

In this chapter, we will be using matrices as tools. You do not need to be an expert in linear algebra, but you should know the basic matrix operations, such as addition, subtraction, multiplication by a scalar, and multiplication by a matrix. For references, see
`https://www.khanacademy.org/math/precalculus/precalc-matrices/Basic_matrix_operations/v/introduction-to-the-matrix`

and

`https://www.khanacademy.org/math/precalculus/precalc-matrices/matrix_multiplication/v/matrix-multiplication-intro`.

4.1 Specifying a Markov Chain

In order to describe a Markov chain, we need a few definitions. We will have a set of *states*,

$$S = \{s_1, s_2, ..., s_n\}, \tag{4.1}$$

which represent possible outcomes for our system.

Our process will start in one of these states and will move from one state to another at each time step. To be precise, if the system is currently in state s_i, it moves to state s_j at the next step with probability p_{ij}. The p_{ij} are called *transition probabilities*. We assume that these probabilities do not depend upon which states the chain was in before the current state.

We will also make a sort of conservation assumption. Namely, we will assume that the sum of probabilities of moving out of a state is equal to one.

Thought Question 43. *If there are n states in the system, how many transition probabilities are there? Can you think of a good way to organize these probabilities?*

Thought Question 44. *If a Markov chain is in state s_i, what notation represents the probability that the system stays in that state?*

The transition probabilities can be arranged in a matrix, called the *transition matrix*, in a way suggested by the notation. Notice that the transition matrix is an $n \times n$ matrix. The probability, p_{ij}, of moving from state s_i to state s_j is the element in the i^{th} row and the j^{th} column.

Thought Question 45. *What can you say about the rows of a transition matrix?*

Problem 46. *They say that in Iceland, if you don't like the weather, just wait 5 minutes. Suppose that if there is a nice day in Reykjavik (the capital of Iceland), they are just as likely to have snow as rain the next day. There are never two nice days in a row. If they have snow or rain, there is a* 50% *chance they will have the same the next day. If there is a change from rain or snow, only half of the time is this a change to a nice day. Use this information to create a Markov chain.*

1. *What are the states of this system? Give them variable names.*
2. *What are the transition probabilities? Arrange them in a transition matrix.*
3. *What information is given by the first row of your transition matrix?*
4. *If it is rainy today, how could you determine the probability that it will be snowy two days from now?*

Problem 47. *We would like to model the college choices of children of legacy students. For simplicity, we will assume that the college choice of a child depends only on the mother's school. Suppose that* 80% *of the children of women from Ripon College go to Ripon and that the rest go to Beloit College. Suppose that* 40% *of the children of women from Beloit College go to Beloit and that the rest split evenly between Ripon College and Lawrence University. Finally, suppose that* 70% *of children of women from Lawrence University go to Lawrence,* 20% *go to Ripon, and* 10% *go to Beloit.*

1. *Create a Markov chain modeling this scenario. What is the transition matrix?*
2. *What is the probability that the grandchild of a Ripon woman attends Ripon College?*

We would like to know what happens if we give our Markov chain initial conditions. In particular, given a certain starting distribution, we will eventually want to study the long-time behavior of a Markov chain.

Definition 48. *A* probability vector *is a row vector whose entries are non-negative and sum to 1.*

If $\vec{u}$ is a probability vector that represents the initial state of a Markov chain, then the i^{th} component of $\vec{u}$ is the probability that the chain starts in state s_i.

Thought Question 49. *Given an initial starting configuration* $\vec{u} = (u_1, ..., u_n)$ *for some Markov chain, how would you calculate the distribution of states after one time step?*

Problem 50.

1. *Let P be the transition matrix as in Problem 46. Suppose that the initial probability vector* $\vec{u} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$*. (How can we interpret this?) Find the distribution of states after 1 day.*

2. *Let P be the transition matrix as in Problem 47. Suppose that you have a collection of women,* 62% *of whom went to Ripon College,* 21% *of whom went to Beloit College, and* 17% *of whom went to Lawrence University. Find the probabilities of their children attending Ripon, Beloit, and Lawrence.*

4.2 Behavior of Markov Chains

We will be interested in using Markov chains to model the distribution of systems over many time steps. As such, it would be convenient if there was a systematic way to do so.

Thought Question 51. *If P is the transition matrix of a Markov chain, let $P^2 = P \times P$, $P^3 = P \times P \times P$, etc. Given your transition matrix, P, in Problem 46, use Sage to find P^2, P^3, P^4, P^5, and P^6. What observations can you make?*

Based on what you have seen above, it seems that considering powers of the transition matrix gives useful information about the behavior of a Markov chain. The next theorem makes that precise.

Theorem 52. *Let P be the transition matrix of a Markov chain.*

1. *The ij^{th} entry, $p_{ij}^{(2)}$, of the matrix P^2 gives the probability that the Markov chain, starting in state s_i, will be in state s_j after 2 steps.*
2. *The ij^{th} entry, $p_{ij}^{(n)}$, of the matrix P^n gives the probability that the Markov chain, starting in state s_i, will be in state s_j after n steps.*

Problem 53. *Prove Theorem 52.1. Hint: how can you succinctly write $p_{ij}^{(2)}$?*

Problem 54 (Challenge)**.** *Prove Theorem 52.2. You may assume you have proven Problem 53.*

Thought Question 55. *Given Theorem 52, can you make any additional observations about powers of the transition matrix in Problem 46?*

If we are given a starting distribution, $\vec{u}$, we saw in the previous section that the probability distribution after one time step is given by $\vec{u}P$. We can generalize that statement to make the following theorem.

Theorem 56. *Let P be the transition matrix of a Markov chain, and let $\vec{u}$ be the probability vector that represents the starting distribution. Then the probability distribution after n steps is given by the vector*

$$\vec{u}^{(n)} = \vec{u}P^n.$$

Sage Problem 57. *Let P be the transition matrix as in Problem 46.*

1. *Suppose that the initial probability vector $\vec{u} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Use Sage to find the distribution of states after 3, 4, 5, and 6 days.*
2. *Now suppose that the initial probability vector $\vec{v} = (\frac{1}{8}, \frac{1}{2}, \frac{3}{8})$. Use Sage to find the distribution of states after 3, 4, 5, and 6 days.*
3. *What observations can you make?*

Sage Problem 58. *The President of the Math Club tells person A his or her intention to run or not to run in the next election. Then A relays the news to B, who in turn relays the message to C, and so forth, always to some new person. We assume there is a probability of 0.3 that a person will change the answer from yes to no when transmitting it to the next person and a probability 0.4 that he or she will change it from no to yes. Let the states of the system be the message–either yes or no.*

1. *What is the transition matrix?*
2. *Suppose the PotMC is equally likely to say yes or no to person A. What are the states of the system after the message passes through 5 people? How should you interpret the output?*

The next two problems will rely on the following information about genetics.[6] The simplest type of inheritance of traits in animals occurs when a trait is governed by a pair of genes, each of which may be of two types, say G and g. An individual may have a GG combination, a Gg combination (which is genetically the same as gG) or a gg combination. An individual is called *dominant* if he or she has GG genes, *recessive* if he or she has gg, and *hybrid* if he or she as Gg.

In the mating of two animals, the offspring inherits one gene of the pair from each parent, and the basic assumption of genetics is that these genes are selected at random, independently of each other. This assumption determines the probability of occurrence of each type of offspring.

Sage Problem 59. *Create a Markov chain that models a continued process of matings. At every generation, an offspring will be chosen at random (assume there is at least one) and mated with a* hybrid*.*

1. *Find the transition matrix.*
2. *If you are equally likely to start with a dominant, recessive, or hybrid individual, find the distribution of states after 10 generations of breeding.*

Sage Problem 60. *Create a Markov chain that models a continued process of matings. At every generation, an offspring will be chosen at random (assume there is at least one) and mated with a* dominant*.*

1. *Find the transition matrix.*
2. *If you are equally likely to start with a dominant, recessive, or hybrid individual, find the distribution of states after 10 generations of breeding.*
3. *How does this differ from Part 2 of the previous question?*

4.3 Regular Markov Chains

An important class of Markov chains are the *regular Markov chains*. A regular Markov chain is one such that the long-range predictions of the model are independent of the starting state.

Thought Question 61. *Give an example of a regular Markov chain. (We have seen at least one already!)*

One way to characterize a regular Markov chain is through the following definition.

Definition 62. *A Markov chain is* regular *if some power of the transition matrix has only positive elements.*

Thought Question 63. *How can you interpret a regular Markov chain? Is it possible to go from any state to any other state? Why or why not?*

Problem 64. *Which of the following are transition matrices for regular Markov chains? Justify your answer in a rigorous manner.*

1. $$P = \begin{pmatrix} .5 & .5 \\ 1 & 0 \end{pmatrix}$$

2. $$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

3. $$P = \begin{pmatrix} 1/3 & 0 & 2/3 \\ 0 & 1 & 0 \\ 0 & 1/5 & 4/5 \end{pmatrix}$$

It may not be clear from the definition why the long-range predictions of a regular Markov chain should be independent of the starting state. The following theorem, which we take without proof, gives the link.[7]

Theorem 65. *Let P be the transition matrix for a regular Markov chain. Then as $n \to \infty$, $P^n \to W$, where W is some limiting matrix having all its rows the same vector $\vec{w}$. Here, $\vec{w}$ is a probability vector with strictly positive entries.*

Thought Question 66. *In your example of a regular Markov chain, what is the limiting matrix W? What are its row vectors $\vec{w}$?*

Given a regular Markov chain, we would like to know how to compute its limiting matrix (why?). The following problem can be used to do just that. First, we give a definition.

Definition 67. *A row vector $\vec{w}$ with the property that $\vec{w}P = \vec{w}$ is called a* fixed row vector *for P.*

Problem 68. *Let P be a regular transition matrix (i.e. a transition matrix for a regular Markov chain). Let $W = \lim_{n\to\infty} P^n$, and let $\vec{w}$ be the row vector of W. Show that $\vec{w}P = \vec{w}$. In other words, $\vec{w}$ is a fixed row vector for P.*

Indeed, it can be shown that any row vector $\vec{v}$ such that $\vec{v}P = \vec{v}$ is a constant multiple of $\vec{w}$. An immediate consequence is that there is only one *probability* vector $\vec{w}$ such that $\vec{w}P = \vec{w}$.

Problem 69.

1. *Recall the Markov chain you constructed in Problem 46. This is a regular Markov chain (why?). Use Problem 68 to calculate the fixed row vector for P.*

2. *Recall the Markov chain you constructed in Problem 47. This is also a regular Markov chain (why?). Use Problem 68 to calculate the fixed row vector for P.*

If we are given an initial starting configuration, the following problem tells us how to find the long-time distribution.

Problem 70. *Suppose that P is the transition matrix for a regular Markov chain, and let $\vec{u}$ be the initial starting configuration. Prove that*

$$\lim_{n\to\infty} \vec{u}P^n = \vec{w},$$

where $\vec{w}$ is the unique fixed probability vector for P.

Definition 71. *We will call the probability vector $\vec{w}$ in Problem 70 the* steady-state distribution *of the regular Markov chain.*

Thought Question 72. *Given a regular Markov chain with transition matrix P, how does the steady-state distribution, $\vec{w}$, relate to W?*

Problem 73. *A certain experiment is believed to be described by a two-state regular Markov chain with the transition matrix P, where*

$$P = \begin{pmatrix} .5 & .5 \\ p & 1-p \end{pmatrix},$$

and the parameter p is not known. When the experiment is performed many times, the chain ends in state one approximately 20% of the time and in state two approximately 80% of the time. Compute a sensible estimate for the unknown parameter p and explain how you found it.

Problem 74. *Consider the Markov chain with transition matrix*

$$P = \begin{pmatrix} 1/2 & 1/3 & 1/6 \\ 3/4 & 0 & 1/4 \\ 0 & 1 & 0 \end{pmatrix}.$$

1. *Show that this is a regular Markov chain.*

2. *The process is started in state 1. Find the probability that it is in state 3 after two steps.*

3. *Find the steady-state distribution of P.*

4.4 Absorbing Markov Chains

Another type of Markov chain that is useful in modeling is an *absorbing Markov chain*.

Definition 75. *A state s_i of a Markov chain is called* absorbing *if it is impossible to leave it. A Markov chain is* absorbing *if it has at least one absorbing state, and if, from every state, it is possible to go to an absorbing state in some number of steps. When a process has reached an absorbing state, we will say that it is* absorbed.

Definition 76. *In an absorbing Markov chain, a state which is not absorbing is said to be* transient.

Thought Question 77. *If s_i is an absorbing state of a Markov chain, what can you say about p_{ii}?*

Thought Question 78. *Can a Markov chain be both absorbing and regular? Why or why not?*

Sage Problem 79 (The Drunkard's Walk)**.** *A man walks along a four-block stretch of Watson St. If he is at corner 1, 2, or 3, then he walks to the right or left with equal probability. He continues until he reaches corner 4, which is a bar, or corner 0, which is his home. If he reaches either home or the bar, he stays there.*

1. *Create a Markov chain with states $0,1,2,3,4$. Draw a diagram if necessary to aid in your construction.*
2. *Are there any absorbing states? If so, which ones?*
3. *Let P be the transition matrix of this Markov chain. Use Sage to compute P^n for some large value of n. What do you observe?*

We will be interested in the long-time behavior of absorbing Markov chains. For example, we would like to know the probability that the process will end in a given absorbing state. We would also like to know how long it takes a process to be absorbed. If a steady state solution exists for such Markov chains, we would like to know that as well.

Thought Question 80. *Consider your Markov chain from Problem 79. Re-order the states so that the transient states come first.*[8] *What does your transition matrix look like?*

Definition 81. *Suppose an absorbing Markov chain has its states ordered in such a way that the transient states come first. If there are r absorbing states and t transient states, the transition matrix will have the following* canonical form*:*

$$\begin{pmatrix} Q_{t\times t} & R_{t\times r} \\ 0_{r\times t} & I_{r\times r} \end{pmatrix},$$

where $Q_{t\times t}$ is a $t\times t$ matrix, $R_{t\times r}$ is a nonzero $t\times r$ matrix, $0_{r\times t}$ is a $r\times t$ zero matrix, and $I_{r\times r}$ is a $r\times r$ identity matrix.

Definition 82. *For an absorbing Markov chain P, the matrix $N = (I_{t\times t} - Q_{t\times t})^{-1}$ is called the* fundamental matrix *for P. The entry n_{ij} of N gives the expected number of times that the process is in the transient state s_j if it started in the transient state s_i.*

Notice here that $(I-Q)^{-1}$ is referring to the matrix inverse of $I-Q$. Recall that if a matrix, A, is invertible, then its inverse, denoted A^{-1}, is the unique matrix such that $AA^{-1} = A^{-1}A = I$.

Theorem 83. *Let t_i be the expected number of steps before the chain is absorbed, given that the chain starts in state s_i, and let $\vec{t}$ be the column vector whose i^{th} entry is t_i. Then*

$$\vec{t} = N\vec{c},$$

where $\vec{c}$ is a column vector all of whose entries are 1.

Problem 84. *Prove Theorem 83.*

We can also determine the long-time behavior of an absorbing Markov chain via the following theorem, which we will take without proof.

Theorem 85. *Let P be an absorbing Markov chain in canonical form.*

1. *If we define B to be $B = (I-Q)^{-1}R$, then entry B_{ij} is the probability that an absorbing chain will be absorbed in the absorbing state s_j if it starts in the transient state s_i.*
2. *Let $L = \lim_{n\to\infty} P^n$. Then*
$$L = \begin{pmatrix} 0_{t\times t} & B_{t\times r} \\ 0_{r\times t} & I_{r\times r} \end{pmatrix}.$$

Sage Problem 86. *Consider your Markov chain from Problem 79 written in canonical form.*

1. *Find the fundamental matrix either by hand or using Sage.*
2. *If we start in state 2, what is the expected number of times before absorption that we will be in state 1? State 2? State 3?*
3. *Find L.*
4. *If we start in state 1, what is the probability of absorption in state 0? State 4?*

Sage Problem 87. *For your Markov chain in Problem 79, assume that the probability that the drunkard will step to the left is $\frac{2}{3}$ and to the right is $\frac{1}{3}$. Find N and B and interpret the results. How does this Markov chain compare to your original one?*

Sage Problem 88. *Assume that a student going to a certain four-year medical school has, each year, a probability q of flunking out, a probability r of having to repeat the year, and a probability p of moving on to the next year. In the fourth year, moving on means graduating.*

1. *Form a Markov chain for this process taking as states* $F, 1, 2, 3, 4$, *and* G*, where* F *stands for flunking out,* G *stands for graduating, and the other states represent the year of study.*

2. *For the case* $q = .1, r = .2$, *and* $p = .7$*, find the time a beginning student can expect to be in the second year. How long should this student expect to be in medical school?*

3. *Find the probability that this beginning student will graduate.*

Sage Problem 89. *This problem will involve a Markov chain that models the process of inbreeding. We start with two animals of opposite sex, mate them, select two of their offspring of opposite sex, and mate those, and so forth. To simplify the example, we will assume that the trait under consideration is independent of sex.*

Let the states of the system be determined by a pair of animals, so that there are 6 states: $s_1 = (GG, GG), s_2 = (GG, Gg), s_3 = (GG, gg), s_4 = (Gg, Gg), s_5 = (Gg, gg), s_6 = (gg, gg)$.

1. *Find the transition matrix for this Markov chain.*

2. *Find N, Nc, and B. Interpret the entries in these matrices.*

3. *Show that the probability of absorption in a state having genes of a particular type is equal to the proportion of genes of that type in the starting state.*

4.5 Project: Chutes and Ladders

In this project, we will use Markov Chains to analyze the popular children's game *Chutes and Ladders*. This project is inspired by [1].

4.5.1 Rules of the game

Chutes and Ladders is played on a board with a 10×10 grid, numbered from 1 to 100, as shown in Figure 4.5.1. At various locations on the board, there are chutes and ladders. Each of these connect a pair of squares.

Players start off the board and take turns by rolling a single 6-sided die and moving the corresponding number of squares. If the completion of a move lands a player at the foot of a ladder, the player instantly climbs to the top of that ladder. Similarly, if a move lands a player at the top of a chute, the player immediately slides to the bottom of the chute. The first player to square 100 wins.

It is important to note that a player must land EXACTLY on square 100 in order to win. If they have a roll that would cause them to move past square 100, they simply do not move during that turn.

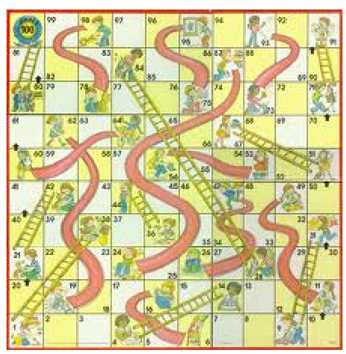


Figure 4.1: A vintage *Chutes and Ladders* game board.

4.5.2 Your assignment

Create a Markov chain to analyze *Chutes and Ladders*. Assume a single player game. (Why is this a reasonable assumption?) Answer any questions that seem useful. In particular, determine the probabilities of various game lengths, in terms of number of turns. What is the most likely game length? (Note: This is an important question, especially if you are playing the game with small children. Given that, theoretically, this game could last forever, you want to make sure that, on average, it doesn't't!)

Time permitting, you might also explore what changes you could make to the game that might influence the average length of play. For example, what would happen if you added more squares, chutes, or ladders? What if you played with a 10-sided die? Answer any questions that seem interesting to you.

You will submit a paper (written in LaTeX) with the details of your analysis. You will not turn in a separate Sage worksheet, so appropriate figures should be included in your paper.

4.6 Instructor's Notes for Chutes and Ladders Project

I like this project because it involves the creation of a large Markov chain, and it requires students to think carefully about how to deal with a large matrix. Typically, my students have handled this in one of two ways: they either enter the matrix by hand in Excel and import as a CSV file into Sage, or they write code in Sage that will populate the transition matrix. Both methods work (although I know which way I would do it!), and to support the first approach, I give them a Sage worksheet with some helpful commands related to CSV files.

Another thing I like about this project is that it came directly from an article ([1]) in *The Mathematical Gazette*. (For a summary of the ideas in this project, see [3].) This formulation of Chutes and Ladders as a Markov chain has prompted a variety of mathematical papers, many with undergraduate student co-authors, including [4], [5], [6]. The last time I taught the course, a group of students and I developed an analysis of this problem, which can be found in *The College Mathematics Journal* [9]. You could have students read any of these articles, if you wanted to emphasize the skill of reading mathematical publications.

I typically give students a week of class time to work on this project. I usually have to encourage them to do a smaller example first, as they often have some difficulty setting up the transition matrix.

Chapter 5

Monte Carlo Simulations

In this chapter, we will be interested in using Sage to simulate the behavior of a real-world situation. The emphasis in this chapter will be on creating simulations, rather than the (very interesting but quite challenging) theory behind the construction and interpretation of these simulations.[9]

We will study *Monte Carlo simulations*, which rely on random numbers to simulate the behavior of a real-world system. We will be interested in creating algorithms to solve both deterministic and stochastic problems.

5.1 Random Number Generators

In order to create a Monte Carlo simulation, we will need to use random number generators.

Thought Question 90. *What is meant by a* random *number? Is it possible for a computer to generate a truly random number?*

Our simulations will require us to use *pseudo-random number generators*. In particular, we will need to use a "good" pseudo-random number generator.

Thought Question 91. *What is meant by a* pseudo-random *number generator?*

Problem 92. *Consider the following algorithm for generating a pseudo-random number.*

- *Start with a 4-digit number. This is the* seed.
- *Square this number to produce an 8-digit number. If the number has fewer than 8 digits, add leading zeroes.*
- *Remove the middle four digits. The remaining 4-digit number is your new seed.*

1. *Start with the seed* 1111. *Use a calculator or a computer to iterate this process 8 times. What do you observe?*
2. *Repeat Part 1 with a different 4-digit seed of your choosing. What happens?*
3. *Modify the third step of this pseudo-random number generator. Instead of removing the middle four digits, remove the first two and the last two digits. Start with the seed* 1234 *and iterate the process 8 times. What happens?*

4. *Repeat Part 3 with the seed* 1001. *What do you observe?*

5. *Is this a "good" pseudo-random number generator? Why or why not?*

The method appearing in Problem 92, Part 3 is called the *middle-square* method and is typically attributed to John von Neumann. He is quoted as saying, "Any one who considers arithmetical methods of producing random digits is, of course, in a state of sin."

Thought Question 93. 1. *What sort of properties should a good pseudo-random number generator have? What sort of properties might bad pseudo-random number generators have?*

2. *Why is it important to use a good pseudo-random number generator when creating simulations?*

We will primarily be interested in generating uniform random numbers; i.e. numbers that are uniformly distributed in some interval. Fortunately for us, Sage has several good pseudo-random number generators available for our use. In the next problem, you will explore ways to use Sage to generate pseudo-random numbers.

Thought Question 94.[10] *There are a variety of ways to generate random data using Sage. Create a notebook that does the following:*

1. *Generate a random floating point number between 0 and 1.*

2. *Simulate rolling a 6-sided die.*

3. *Generate a random odd number between 1000 and 3000.*

4. *Generate a random floating point number between 23 and 47.*

5. *Create a list of 500 randomly generated days of the week.*

In our simulations, we will want to generate lists of pseudo-random numbers and determine properties about them. The following problems will explore this.

Sage Problem 95. *In this problem, you will work with products of randomly generated numbers.*

1. *Write a function in Sage that reads in 3 integers and calculates their product.*

2. *Use your function to create a list of 500 numbers, each of which is the product of 3 random integers between 1 and 10.*

3. *Display the frequency of each value attained in a histogram.*

Sage Problem 96.

1. *Write a function in Sage, called "Roll dice" that simulates rolling two 6-sided die.*

2. *Using your function, create a program that allows you to generate 50 dice rolls.*

3. *Determine the frequency of each value rolled. Plot your results in a histogram.*

5.2 Using Monte Carlo Simulations to Solve Deterministic Problems

In this section, we will see that Monte Carlo methods can be used to examine deterministic problems. In particular, we will use Monte Carlo simulations to estimate the value of definite integrals.

Definition 97. *A* Monte Carlo simulation *is a computational algorithm that uses random numbers to simulate the behavior of a situation.*

A deterministic model is one in which no randomness is involved in the behavior of the system. In other words, a deterministic model will always produce the same results for a given input or initial condition.

Thought Question 98. *Can you think of an example of a deterministic system or model that we have seen in this course?*

We can use Monte Carlo simulations to study such systems. We will use Monte Carlo methods to approximate the area under a graph of a non-negative function, $f(x)$, on the interval $[a,b]$. We will generate many random ordered pairs and determine how many lie under the graph of the function. This should help us determine an estimate for the area under the curve. The algorithm is as follows:

Input: The total number of random points, N, to be found; the non-negative function, $f(x)$; the interval $[a,b]$ containing x-values; the interval $[0,M]$ containing y-values, where $M > \max f(x)$ on $[a,b]$.

Output: The approximate area under $y = f(x)$ on the interval $[a,b]$.

Step 1: Specify the function, $f(x)$, and set a counter variable to 0.

Step 2: Calculate random coordinates (x_i, y_i) in the rectangular region:

$$a \leq x_i \leq b, \qquad 0 \leq y_i \leq M.$$

Step 3: For i from 1 to N, do steps 4 and 5.

Step 4: Calculate $f(x_i)$.

Step 5: Compare y_i and $f(x_i)$. If $y_i \leq f(x_i)$, then increment the counter by 1. Otherwise, do not increment the counter.

Step 6: Estimate the area by $A = M \times (b-a) \times \frac{counter}{N}$.

Thought Question 99. *Write a short paragraph describing this algorithm in your own words. How does it work?*

Sage Problem 100. *In this problem, we would like to use a Monte Carlo simulation to approximate the area under the curve $y = x^3$ on the interval $[0,2]$.*

1. *Use the algorithm described above to write a Sage program to approximate this area. Include a plot of the curve and the random points generated on the same set of axes.*
2. *Using calculus, find the exact area under the curve $y = x^3$ on $[0,2]$. How does this compare to your estimate?*

Sage Problem 101. *Use a Monte Carlo simulation to approximate the area under the curve $y = \frac{2}{\sqrt{\pi}} e^{-x^2}$ on $[0,5]$. Include a plot of the curve and the random points generated on the same set of axes.*

Notice that this function has no closed form antiderivative. Its antiderivative is known as the error function, erf(x). In other words,

$$erf(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

Sage Problem 102. *Using an algorithm similar to those in the previous two problems, create a Monte Carlo simulation to approximate the value of π. (Hint: Is there a curve with area underneath it equal to π?)*

Sage Problem 103. *Modify the algorithm used above to compute the volume in the first octant under the surface $x^2 + y^2 + z^2 = 1$.*

5.3 Monte Carlo Methods and Stochastic Events

A stochastic event is one that involves some degree of randomness. In modeling a stochastic system, the same initial data may not lead to the same behavior of the system. Perhaps unsurprisingly, we can use Monte Carlo simulations to study such systems.[11]

Thought Question 104. *Can you think of a stochastic model that we have studied in this class?*

We can use Sage to simulate probabilistic behavior. We will often want to write functions to use in simulations.

Sage Problem 105. *In this problem, we would like to write a function that reads in the number of rolls of a fair die to make (call this n). The function will output the results of each roll into a list.*

1. *Write a short description of the algorithm that you will use to create your function. What is the input for your function? What is the output?*
2. *Write a Sage function that does this.*
3. *Run your procedure for $n = 10, 100, 1000$. For each value of n, create a histogram or a bar chart that displays the number of occurrences of each die value.*
4. *Calculate the probability of obtaining each value, $1-6$, of the die. What happens as n gets large?*

Sage Problem 106. *In this problem, we would like to simulate the following simple game. Players A and B each roll a fair die. The one who gets the larger number wins the game. If the players get the same number, they tie.*

1. *Write a short description of the algorithm that you will use to create your game. Incorporate at least two functions, including the one from Problem 105.*
2. *Write a program that simulates the game.*
3. *Repeat the game 1000 times. Count how many times each player wins and how many games end in a tie.*
4. *Display the results of your game in an appropriate graphical format.*

The following two problems will allow us to compare results obtained from Monte Carlo simulations to those obtained from Markov Chains.

Sage Problem 107 ("The Drunkard's Walk" Revisited)**.** *Suppose that drunk person leaves a bar in the middle of a very long street. The street is so long that there is one block for every integer. Assume the bar is located at block zero.*

1. *Write a Sage function that simulates the drunkard's walk. Assume he is equally likely to move left or right at every block. The procedure should calculate the drunkard's position after traveling n blocks, where n is a positive integer to be entered by the user.*

2. *Create a list of 100 walks that are each 20 blocks long. Do a tally to see how many walks end on each block. Plot your data in a reasonable fashion.*

3. *Based on your simulation, estimate the probability that the drunkard will end on block 10 after a 20 block walk.*

Sage Problem 108 ("The Drunkard's Walk Revisited" Revisited)**.** *Recall the drunkard's walk in Problem 79.*

1. *Write a Sage function that simulates this problem. You should include the details of that problem (5 blocks, with blocks 0 and 4 always ending the walk). Assume that he is equally likely to move left or right at each step.*

2. *Using your simulation, estimate the probability that the drunkard will end in state 0, if he starts in state 2. Estimate the probability that he will end in state 4, if he starts in state 2. How does this compare to your work in Problem 79?*

3. *Estimate the probability of absorption in states 0 and 4, given that he always starts in state 1.*

4. *Estimate the probability of absorption in states 0 and 4, given that he always starts in state 1.*

5. *Modify your simulation so that the drunkard has a $\frac{2}{3}$ chance of moving left and a $\frac{1}{3}$ chance of moving right. Find the probabilities of absorption in 0 and 4 for each starting state.*

5.4 Project: Election Imperfection

The Math Club is planning to have elections to determine its President (PotMC). Unfortunately, at least one member of the club, Stu Born, has studied some voting theory and knows that it is possible for the same set of data to yield different winners under different voting methods.

For example, suppose that Candidates Euler, Germain, and Mirzakhani are running in an election. Suppose that 5 voters prefer Euler, then Mirzakhani, then Germain, suppose that 4 voters prefer Germain, then Mirzakhani, then Euler, and suppose that 1 voter prefers Mirzakhani, then Germain, then Euler. Under the plurality method of voting, Euler would win with 5 first place voters. However, under the Borda count method, Mirzakhani would win.

Because of this, Stu argues, the Math Club should reject the notion of holding fair elections. "Basically," he reasons, "if elections *can* produce different outcomes with the same data, then they *will* tend to produce different outcomes, so results obtained from voting are meaningless." To fix this issue, Stu claims that he should just be appointed lifetime President (LPotMC).

5.4.1 Your assignment

- Read the "Voting and Elections" Column at the AMS (`http://www.ams.org/samplings/feature-column/fcarc-voting-introduction`).
- Develop a Monte Carlo simulation that calculates the Condorcet efficiency (i.e. frequency with which the Condorcet winner is elected **should one exist**) for at least three different voting systems.
- Your program should allow for a variable number of candidates and voters.
- You should implement an impartial-culture (totally random preference orders), a bipolar-culture (first third has one preference order, second third has reverse preference order of the first, and last third have impartial-culture), and a perturbation of impartial-culture towards unipolarity (5-10% of the voters have a fixed preference order and the rest have impartial-culture).

Based on the results of your simulation, you should decide whether Stu is correct, or whether it is possible to get coherent results from voting. Further, you should make a recommendation to the Math Club about which voting method they should use. Write a formal report in LaTeXdetailing your findings.

5.5 Instructor's Notes for Election Project

This project is based upon a conversation in Political Science between William Riker in [16] and Gerry Mackie in [12]. Riker claims that since voting theory shows us that no voting system is perfect (Arrow's theorem), that every voting system is susceptible to tactical voting (Gibbard-Satterthwaite theorem), and that there can be inconsistencies between different voting systems on the same set of data, we must reject any form of democracy that utilizes voting as a mechanism in the democratic process. Basically, one of his arguments is similar to Stu Born's.

> The moral and prudential standoff among methods would not in itself occasion difficulty for democratic theory if 'most of the time' most methods led to the same social choice from a given profile of individual values...But it seems a safe conjecture that, if such a comparison [of commonly discussed methods] were made, the proportion of social profiles from which all the compared methods produced identical results would indeed be tiny. [16]

In [12], Mackie responds by providing evidence that it is not the case. In Chapter 3, he presents the results of a number of simulations done regarding consistency of voting methods and shows that, in fact, most reasonable voting methods tend to agree under pretty mild conditions (like a slight perturbation away from impartial-culture). This project is based upon Mackie's text as well as [15].

This project requires a somewhat high degree of programming proficiency (working with lists and functions at an intermediate level), so I assign this project to the students in class who have programming experience. It also is a longer project, probably necessitating 2 weeks of class time.

5.6 Project: Population Modeling

Monte Carlo simulations can be powerful tools in real-world modeling, because real-world processes often have a degree of randomness to them. This project will involve using Monte Carlo simulations to study the bobcat population in Florida. This project is adapted from [14].

The bobcat is an endangered species, and population management of this species is important. In this project, you will simulate the bobcat population and provide recommendations about management strategies.

5.6.1 Your assignment

You should create a model of the bobcat population in a region in Florida. You may use the information provided below, but you should remember that growth rates really are stochastic.

- Based upon growth rate data from [7], annual growth rates for bobcats in Florida seem to depend largely on environmental conditions. On average, they seem to be about 0.1676 (best conditions), 0.00549 (medium conditions), and -0.04500 (worst conditions).
- Management strategies can include hunting and/or adding some number or percentage of the bobcat population each year.
- Florida has extreme weather events occur with some frequency.
- There may be an outbreak of a disease that is harmful to bobcats.

You should provide an analysis of three different regions: one with the best conditions, one with medium condition, and one with the worst conditions for bobcat survival. You may assume an initial population of 100 bobcats for each model. You should develop management strategies for each region that keep the bobcat population between 50 and 200 for the next 100 years. You will submit your analysis in a LaTeXdocument.

5.7 Instructor's Notes for Population Project

This project combines difference equations with Monte Carlo simulations to provide a more realistic population growth model. It also allows students to see the power of using mathematical modeling to experiment with different management strategies and to see how mathematics could be used to influence policy decisions.

This project takes about one week of class time to complete for students with little programming experience. Students with strong programming backgrounds could likely complete it much more quickly.

Chapter 6

Linear programming

In this chapter, we will be interested in solving *linear programming problems*, which are problems that optimize a linear function subject to certain linear constraints. These problems have a wide variety of applications. This chapter will follow the treatment in [2] , [10], and [17].

6.1 Formulating Linear Programming Problems

To begin, we would like to understand what types of scenarios can be formulated as linear programming problems. We will make the following definitions.

Definition 109. *Let $F(x_1, \ldots, x_n)$ be a linear function; i.e.*

$$F(x_1, \ldots, x_n) = a_1 x_1 + \cdots + a_n x_n, \tag{6.1}$$

where a_i are known parameters. We call the x_i the decision variables*, while the function F is called the* objective function*.*

In a linear programming problem, we would like to find values of the decision variables that will maximize or minimize F, subject to a set of *linear* constraints, which will typically be of the form

$$m_{i1} x_1 + \cdots + m_{in} x_n \leq b_i, \tag{6.2}$$

for $i = 1, \ldots, m$. We may interpret this system of constraints geometrically as defining a region such that, at every point in the region, the constraints are all simultaneously satisfied. This region will be called the *feasible region*.

Thought Question 110. *Suppose you are given the following constraints:*

$$\begin{aligned} 2x_1 + 4x_2 &\leq 1400, \\ 4x_1 + 3x_2 &\leq 1500, \\ x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

Draw the feasible region indicated by these constraints.

Thought Question 111. *Formulate the linear programming problem given by Equations 6.1 and 6.2 in terms of vector and matrix notation. Recall that the transpose of a row vector is a column vector and is denoted x^T. If $(x_1, \ldots, x_n)$ is a row vector, then*

$$x^T = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Suppose we are considering a linear programming problem with objective function $F(x_1, x_2)$. Then the feasible region is a region in $\mathbb{R}^2$. We can solve such a problem using the following idea.

Definition 112. *For any real number c, $F(x_1, x_2) = c$ is called an* isoprofit line. *For all points on this line, the "profit" (as measured by F) is the same.*

If we want to maximize F subject to constraints, we must find the isoprofit line with the largest value of c such that (x_1, x_2) is a point in the feasible region.

Problem 113.

1. *In the case of 2 decision variables, describe as precisely as possible the shape of a feasible region.*
2. *Why does the equation $F(x_1, x_2) = c$ as in Definition 112 represent the equation of a line (for a fixed c)? What is its slope? What is its y-intercept?*
3. *Draw a picture that shows how an isoprofit line could be used to maximize a function within a feasible region.*

Problem 114. *A wine maker would like to decide how many bottles of red wine and how many bottles of white wine to produce in order to maximize his profit. Given his expertise is in red wine making, he can sell a bottle of red wine for $12, whereas he can only charge $7 for a bottle of white wine.*

Aging wine in wooden or glass-lined vats is an important component of the production process, but due to limited space, the wine must be aged for a limited time. The wine maker has determined that red wine should be aged two years per bottle, and white wine should be aged one year per bottle. His facilities allow that each batch is limited to 10,000 bottle-years (5 bottles of red and 3 bottles of white require a total of 13 bottle-years).

Also, the volume of grapes that may be processed is limited, and it takes 3 gallons of grapes to make a bottle of red wine and two gallons of grapes to make a bottle of white wine. Furthermore, the winery can only process a total of 16,000 gallons of grapes for each batch.

1. *Formulate this problem as a linear programming problem; i.e. find the objective function and the constraint equations. Define what the decision variables represent.*
2. *Graph the feasible region.*
3. *Use isoprofit lines to solve this problem. What is the wine maker's maximum profit?*

Problem 115. *A child with a new 29 gallon fish tank asks her dad to put as many fish as possible in the tank. She wants only two varieties: the big orange fish (Gouramis) and the small stripy fish (Zebra Danios). Sensing that too many fish is not a good thing, the dad asks the pet shop owner how many fish can go into a tank. The answer was more complex than anticipated. "You can put one inch of fish in per gallon of water." A full grown Gourami is two inches long, while a Danio is one inch long.*

The dad decides that he would prefer not to go broke buying fish food, and thus he wants to limit the tank to 50 grams of food per day. Danios are very active fish and each require 4 grams per day, while the slower Gouramis each eat only 2 grams per day.

Finally, the pet shop owner points out that Danios need to live in schools of at least 5 fish or they don't do well. The little girl points out that she must have at least two Gouramis, since they are known to kiss.

1. *Help this poor dad by formulating this scenario as a linear programming problem.*
2. *Graph the feasible region.*
3. *Use isoprofit lines to solve this problem. What is the maximum number of fish the girl can have in her tank?*

Problem 116. *A small business wants to know how many of two types of high-speed computer chips, type A and type B, to manufacture weekly in order to maximize its profits. The company reports a profit of \$140 for each type A chip and \$120 for each type B chip sold. Chip A takes 2 hours to assemble, whereas Chip B takes 4 hours to assemble. The assembly line can run for 1400 hours per week. Chip A takes 4 hours to install, while Chip B takes 3 hours to install. There are 1500 hours available each week for installation.*

1. *Formulate this scenario as a linear programming problem.*
2. *Graph the feasible region.*
3. *Use isoprofit lines to solve this problem. How many of each type of chip should the business make in order to maximize their profits?*

Problem 117. *In Problems 114, 115, and 116, the optimal solution to the problem occurred at a vertex of the feasible region. If F depends on 2 variables x_1 and x_2, will this always be the case? Why or why not?*

6.2 Sensitivity Analysis

In a linear programming problem, it is often the case that one does not precisely know the values of the coefficients. As such, *sensitivity analysis*, or the behavior of the system with respect to changes in parameters, is of great importance.

Thought Question 118. *For a linear programming problem, what parameters could we change? What would these changes represent?*

One aspect of the problem we could consider is price sensitivity.

Problem 119. *Consider Problem 114. Suppose that instead of charging \$7 for a bottle of white wine, the wine maker would like to know how the price of a bottle of white wine affects the optimal solution.*

1. *Reformulate your objective function to reflect an unknown cost of white wine.*
2. *Solve the problem using isoprofit lines. Do your answers depend on the amount charged for a bottle of white wine? If so, how?*

We can also consider resource sensitivity.

Problem 120. *Again consider Problem 114, with all parameters as given in that problem. Now suppose that the number of gallons of grapes the winery can process per batch is unknown, as is the number of bottle-years that each batch can be aged.*

1. *Reformulate your constraints to reflect these unknowns.*
2. *How do the values of these unknowns affect possible solutions?*

Finally, we can see how sensitive the problem is to changes in constraint coefficients.

Problem 121. *We will once again start with Problem 114. Suppose the amount of time that we age a bottle of red wine is allowed to vary.*

1. *Reformulate your constraints to reflect this unknown.*
2. *If the amount of aging time is changed to 1.95 years, how does this affect your solution?*

Problem 122. *Consider Problem 115. Conduct a sensitivity analysis of your choosing on this problem.*

6.3 Built-in Functions for Linear Programming

Graphical methods, such as the ones used in the previous two sections, work well for solving linear programming problems that have only two variables. However, most real-world linear programming problems will have many more variables, and solving these problems by hand is **difficult**. Fortunately, there are solvers in many programming languages that will solve linear programming problems for you. We will use the "Mixed Integer Linear Program" class in Sage.[12]

As you know at this point in the class, learning how to use the computer in mathematical modeling requires you to learn about new programs or commands. Often, you will need to read the documentation and experiment with commands in order to understand how they work. You will get a chance to practice these skills in this section. Before you attempt the problems in this section, you should *carefully* read the documentation about the "Mixed Integer Linear Program" class, which can be found here:
`http://doc.sagemath.org/html/en/reference/numerical/sage/numerical/mip.html#sage.numerical.mip.MixedIntegerLinearProgram`.

Sage Problem 123. *Your solution to this problem can take the form of an example using the Mixed Integer Linear Program (MILP) class, with the answers to the following questions written as comments.*

1. *How do you define an instance of the MILP class? How do you tell Sage you want to do a maximization problem? A minimization problem?*
2. *How do you add variables to the linear program? What inputs does this function take?*
3. *How do you add a constraint? What inputs does this function take?*
4. *How do you define the objective function? What inputs does this function take?*
5. *How do you display all of the information contained within your linear program?*
6. *How do you solve the linear program? What value(s) get returned?*
7. *If not all of the values you would like for your solution are returned with your command in the previous question, how do you get the other values you would like?*

Sage Problem 124. *Use the MILP class to solve Problem 114.*

Sage Problem 125. *Use the MILP class to solve Problem 115.*

Sage Problem 126. *Adventure Time offers one-week summer vacations in the Rocky Mountains during the month of August. The package includes round-trip transportation and a week's accommodations at Rocky Mountain Lodge. The Lodge gives a discount to Adventure Time if they rent two- or three-week blocks of condos, giving a rent of \$1000 per condo for a two-week period and \$1300 per condo for a three-week period.*

Adventure Time expects to need at least the following number of condos: 30 for the first week, 42 for the second week, 21 for the third week, and 32 for the fourth week.

How many condos should Adventure Time rent for two weeks, and how many should be rented for three weeks, to meet the needed number and to minimize Adventure Time's rental costs? Use the MILP class to solve this problem.

Sage Problem 127. *Suppose you own a restaurant, called "Mathematical Meals," that is open 7 days a week. Based on your past experience, the smallest number of workers you need on a particular day is given as follows: Every worker works 5 consecutive days and*

Day	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Number	14	13	15	16	19	18	11

then takes 2 days off, repeating this pattern indefinitely. How should you schedule workers in order to minimize the total number of workers required to staff Mathematical Meals?

Sage Problem 128. *Suppose you wanted to find the "perfect" diet. Ideally, your diet would meet or exceed basic nutritional requirements, be inexpensive, have variety, and be tasty. We can use linear programming to find such a diet.*

Suppose the only foods in the world are as follows:

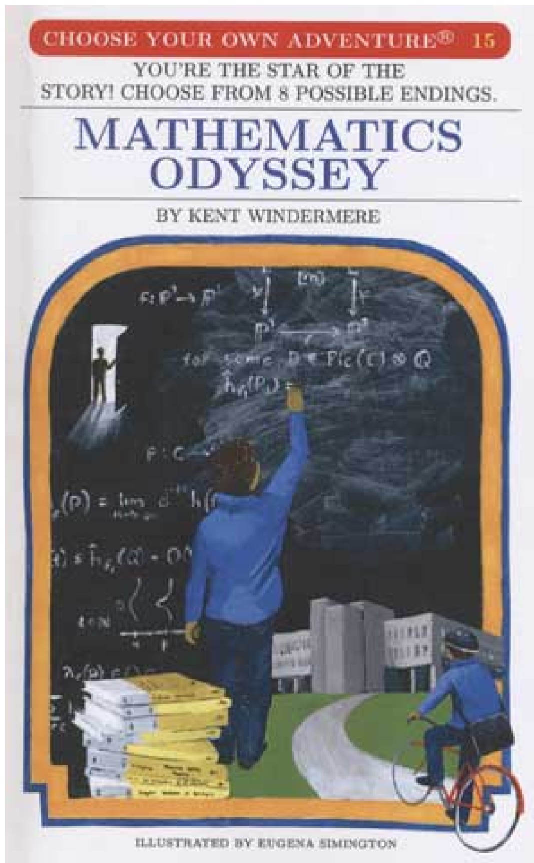
Food	Serving Size	KCal	Protein (g)	Calcium (mg)	Cents/serving	Max servings/day
Oatmeal	28 g	110	4	2	3	4
Chicken	100g	205	32	12	24	3
Eggs	2 large	160	13	54	13	2
Whole milk	237 mL	160	8	285	9	8
Cherry Pie	170g	420	4	22	20	2
Pork and Beans	260 g	260	14	80	19	2

Suppose a satisfactory diet has at least 2000KCal, 55 g of protein, and 800 mg of calcium. The diet should have no more than the maximum number of servings per day of each food item listed in the table. What is the lowest cost satisfactory diet subject to these constraints?

Sage Problem 129. *Design your own linear programming problem that has a minimum of five variables and seven constraints. Your example should be based on a real-world scenario that is relevant to your life in some way. You should solve your linear program using the MILP class.*

6.4 Project: Choose your own adventure

Choose your own scenario to model as a linear programming problem. Be creative. Your written solution should include the use of the simplex method, but you can use Sage. Answer any questions that may be interesting or relevant to your model.



6.5 Instructor's Notes for Linear Programming Project

At this point of the semester, the students are working on their final modeling project, so I do not do a project for linear programming. However, I think it would be worthwhile to have them develop their own model, especially if they can find a real-world application using actual data.

Appendix A

Skills Checks

A.1 Chapter 2

Sage Problems

For these problems, you should submit a single Sage worksheet, with explanations written in Markdown cells.

1. Consider the following difference equation:

$$a(n+1) = (a(n) - .2) * a(n) + a(n).$$

   Use Sage to plot the solution on the interval $[0, 10]$ with initial condition $a(0) = 0.1$.

2. Find the fixed point(s) of the following system:

$$\begin{aligned} x(n) &= 2x(n-1) + 2y(n-1) + 15 \\ y(n) &= -x(n-1) + 3y(n-1) - 7. \end{aligned}$$

   Use Sage to draw a phase plane portrait. Would you say the fixed point(s) are stable or unstable?

LaTeXProblems

For these problems, you should submit a single LaTeXdocument, with explanations clearly written.

1. Find the fixed points of the following difference equation and test for stability.

$$x(n) = (x(n-1))^2 + .7x(n-1) + .02$$

2. Construct a difference equation that has a stable fixed point at $x = 1$ and an unstable fixed point at $x = 3$.

A.2 Chapter 3

All of these problems should be written up in LaTeX.

1. Consider the game of tennis when *deuce* is reached. If a player wins the next point, she has *advantage*. On the following point, she either wins the game or the game returns to *deuce*. Assume that for any point, player A has probability p of winning the point.
   (a) Set this up as a Markov chain. (Hint: you should have 5 states.)
   (b) For $p = 0.6$, find the absorption probabilities.
   (c) At deuce, find the expected duration of the game and the probability that B will win.

2. Assume that a person's profession can be classified as professional, skilled laborer, or unskilled laborer. Assume that, of the children of professionals, 80% are professional, 10% are skilled laborers, and 10% are unskilled laborers. In the case of children of skilled laborers, 60% are skilled laborers, 20% are professional, and 20% are unskilled. Finally, in the case of unskilled laborers, 50% of the children are unskilled laborers, and 25% each are in the other two categories. Assume that every person has at least one child.
   (a) Form a Markov chain for this scenario.
   (b) Find the probability that a grandchild of an unskilled laborer is a professional.
   (c) Find the long-time probability distribution of this Markov chain. Prove that your answer gives the long-time behavior (i.e. don't rely on Sage).

3. Consider a Markov chain with the following transition matrix:
$$P = \begin{pmatrix} .2 & .1 & .5 & .2 \\ 0 & 1 & 0 & 0 \\ .3 & .2 & .25 & .25 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$
   (a) What type of Markov chain is this?
   (b) Find the fundamental matrix. Give a complete interpretation of this matrix.
   (c) Use an appropriate theorem to find the matrix that gives the long-time behavior of this system. Give a complete interpretation of the result. (You should think carefully about what would constitute a *complete* interpretation. What kinds of questions can you answer?)

4. Let P be the transition matrix for a regular Markov chain and $\vec{v}$ an arbitrary probability vector. Prove that
$$\lim_{n\to\infty} \vec{v}P^n = \vec{w},$$
where $\vec{w}$ is the unique fixed probability vector for P.

A.3 Chapter 4

You should submit a single Sage worksheet that is appropriately commented. It should include your explanations in complete sentences written in Markdown cells.

1. In this problem, we will use the curve $y = -x^2 + 5x - \frac{1}{4}$.
   (a) Use a Monte Carlo simulation to approximate the area under this curve on the interval $[1,4]$. Compare your value to the exact value.
   (b) Graph the sampled points and the curve on the same set of axes. You should display the set of points below the curve in one color and the points above the curve in a different color. The curve should be in a third color.
2. Recall Problem 4.3.3 from your notes that simulated rolling a fair die n times.
   (a) Modify the problem to write a function that allows the user to input the number of rolls, n, as well as the number of sides, m, the die has.
   (b) Write a separate function that calculates the probability of rolling each of the values $1,\ldots,m$.
   (c) Write a short paragraph describing the algorithms you used for your functions.
   (d) Run your functions for 3 different m values, letting $n = 1000$ each time. You must call the functions properly.
   (e) Analyze the output of each trial using a counter and in an appropriate graphical format.

A.4 Chapter 5

*For this problem, you should submit a Sage worksheet that is appropriately commented. It should include your explanations in complete sentences written in **Markdown cells**.*

1. The Superbowl Advertising Agency (SAA) wishes to plan an advertising campaign in three different media–television, radio, and magazines. The purpose or goal is to reach as many potential customers as possible. Results of a marketing study are shown in the following table:

	Daytime TV	Prime-time TV	Radio	Magazine
Cost of advertising unit	\$40,000	\$75,000	\$30,000	\$15,000
Number of potential customers reached per unit	400,000	900,000	500,000	200,000
Number of female customers reached per unit	300,000	400,000	200,000	100,000

The company does not want to spend more than \$800,000 on advertising. It further requires the following:

(a) At least 2 million exposures take place among women.

(b) TV advertising be limited to \$500,000.

(c) At least two advertising units be bought on daytime TV and four units on prime-time TV.

(d) The number of radio and magazine advertisement units should each be between 5 and 10 units.

Set this up as a linear programming problem and use Sage to solve. Fully explain your problem and your solution.

For this problem, you should submit a LaTeX document. You may draw your figures by hand or in Sage and insert them into your file.

2. Let $F(x,y) = x + y$, where x and y are subject to the following constraints:

$$\begin{aligned} 4x+2y &\leq c_1 \\ 2x+3y &\leq c_2 \\ x,y &\geq 0, \end{aligned}$$

where c_1, c_2 are positive constants. Use isoprofit lines to maximize F over the feasible region. How does your answer depend on the choice of parameters? You should include several carefully drawn pictures. (Hint: Your answer should have three different cases.)

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Notes to the Instructor

[1]My students seem to struggle initially with the idea of recursion, probably because for many of them, this is the first time to see recurrence relations. Typically after the first day of presentations, once the students see recursion, they fare much better with the material. So you might consider introducing the concept in advance. (I do not, but I don't mind spending a day in class discovering that idea.)

[2]This is a particularly useful problem, both in terms of the model itself and in terms of the demonstration of a chaotic system (in part 4). It will be referred to later in the chapter and can be used as a building block piece in the project.

[3]I usually skip this section, as it is more theoretical and less focused on applications. However, depending on the interests of your students, you might include it.

[4]The mass action assumption governing interactions between two populations is important to highlight. Students should (or could) use that type of term in their Chapter 2 project.

[5]It is a little finicky to plot this system. Students often have the difficulty that their solutions blow-up rapidly. You have to use good choices of parameters. I ask the students to really think about what the parameters mean, but it is hard to come up with good choices without data.

The following system gave a nice plot:

$$\begin{aligned}
P(n+1) &= (1-0.1)P(n)+0.005P(n)N(n)\\
N(n+1) &= (1+0.05)N(n)-0.005P(n)N(n)\\
P(0) &= 5\\
N(0) &= 50.
\end{aligned}$$

[6]The first example is an example of a regular Markov chain. The second example is an absorbing chain. The difference between the two can be previewed in Part 2 of each question.

[7]I choose to just cite this theorem (as I do several places in the notes) because the proofs are too complicated for the students in my course. However, you could easily modify the notes to include these proofs.

[8]The remainder of this section requires the manipulation of matrices. While much of this can be done by hand, it is useful for the students to learn how to do it in Sage.

[9]This chapter relies heavily on programming in Sage. I usually have a range of programming expertise among my students. It is highly advantageous to have students show

their code during presentations, so they can demonstrate how different commands work for their classmates.

[10]This is a thought question because we have done this material on the Chapter 4 Sage Day. If you do not spend a day doing preparatory Sage work, you might want to assign this as a regular problem.

[11]This section is largely here to teach students how to write functions in Sage. The last time I taught the course, half of the students had significant programming experience and half had only what we had done in this course. I allowed the first half to skip this section and move onto the Election Imperfection Project, while the latter half worked through this section and then completed the Population Modeling Project.

[12]If you are not using Sage, you will need to reformulate this section for your own programming language.